

Addressing Illiteracy of Vision-Language Model in Underrepresented Language Through Image-Text Mix Augmentation Scheme

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Abstract

Recently, open-source large Vision-Language Models (VLMs) have progressed toward achieving performance comparable to closed-source VLMs. However, open-source VLMs struggle to recognize unfamiliar texts depicted in the images, where the texts are written in underrepresented languages like Korean. This illiteracy problem is primarily due to insufficient training data for the underrepresented languages. To address this problem, we propose a novel augmentation scheme that generates large-scale image data for the underrepresented languages with minimal manual annotations. Our scheme synthetically combines a text image depicting words or sentences with a template image containing textual contexts, such as a receipt, a sign, a book, and a product label. Specifically, the text image is cut and pasted into a patch of the template image to generate a synthetic image, which is labeled with the corresponding texts in the text image. The synthetic images can reflect underrepresented languages. We also adopt diverse text types, mixing strategies, and prompt designs to express various real-world contexts. Therefore, fine-tuning a VLM with our synthetic data can enhance its ability to generalize to real-world text recognition tasks. Experimental results demonstrate the effectiveness of our scheme, showing a significant performance improvement in text recognition.

1. Introduction

The advancement of Large Language Models (LLMs) and Vision-Language Models (VLMs) has revolutionized the field of natural language processing and multimodal understanding [16, 13, 12, 18, 11, 17]. LLMs focus on processing and generating human language, while VLMs extend these capabilities by integrating visual inputs, enabling models to handle tasks that involve both texts and images. Multimodal Large Language Models [6, 2] take this a step further by utilizing multiple types of data such as text, images, audio, etc., allowing for context-aware understand-

ing across different modalities. These models have demonstrated remarkable capabilities in various tasks, from language generation to speech-to-speech.

In recent years, open-source VLMs [12, 3, 18, 11, 17] progress toward achieving performance comparable to closed-source ones such as Gemini [16] and GPT-4o [13]. The open-source VLMs can offer significant potential applications, leading to widespread adoption in both academia and industry. However, the training data of the open-source VLMs are heavily biased towards images representing Chinese and English texts, where resources are abundant. Therefore, the open-source VLMs exhibit limitations in recognizing texts from images written in underrepresented languages like Korean, Arabic, and Turkish, due to a scarcity of training data. We define this problem as “illiteracy” of VLMs and aim to address the illiteracy problem. Intuitively, humans learn how to read by visually encountering numerous texts. Accordingly, we assume that the VLMs can enhance their text recognition capabilities by learning from large-scale text images.

To this end, we propose a novel augmentation scheme called Image-Text Mix Augmentation that generates a large-scale synthetic dataset for underrepresented languages with minimal human annotations. This method integrates a text image (e.g. words or sentences) into a template image representing common textual contexts (e.g. receipts, signs, books, and product labels). Specifically, the text image is cut and pasted into a designated patch of the template image to generate a synthetic image, which is labeled with the corresponding texts in the text image. This synthetic data generation process creates images containing underrepresented languages. We also adopt diverse text types, mixing strategies, and prompt designs to express diverse real-world scenarios, which can improve the VLM’s generalization capabilities on text recognition tasks through fine-tuning.

Among the underrepresented languages, the Korean language presents a unique challenge. Recent open-source VLMs like MiniCPM-V [17] often struggle to understand Korean texts while showing strong performance in English and Chinese texts. This discrepancy highlights the neces-

sity of a large-scale Korean text dataset. Therefore, we propose a new dataset called a Korean Image-Text Mix dataset, which contains synthetic images generated from the proposed augmentation scheme. To reflect real-world scenarios, we construct various types for the synthetic images in the dataset: 1) single-word frontal, 2) multi-word frontal, 3) single-word skewed, and 4) multi-sentence frontal types. Using our training dataset, we fine-tune the vision language model, MiniCPM-V, through low-rank adaptation [8]. The vision language model fine-tuned on our training dataset is verified in-depth with ablation studies and achieves improved performance on our test dataset, showing the effectiveness of our scheme. Also, the fine-tuned model shows promising results in the public real-world dataset in zero-shot evaluation that does not undergo additional training.

Our main contributions are as follows:

- For large-scale realistic datasets containing underrepresented languages, we propose a novel augmentation scheme that generates a synthetic image dataset, where text images are cut and pasted into patches of template images, and labels are the corresponding texts.
- To address the inadequate training problem of existing vision language models due to the scarcity of Korean datasets, we propose a new dataset called a Korean Image-Text Mix dataset, which constructs four types of synthetic images from our scheme.
- The vision language model fine-tuned on our training dataset is verified in-depth with ablation studies and achieves improved performance on our test dataset. Also, the fine-tuned model shows promising results on the public real-world dataset in zero-shot evaluation.

2. Related Work

2.1. Data Augmentation

Data augmentation has long been a fundamental technique to enhance the robustness and generalization of deep learning models, particularly in tasks involving image processing. Traditional methods such as random cropping, flipping, and rotation expand datasets by introducing basic variations. Also, advanced augmentation techniques have emerged, such as Mixup [20], Cutout [4], and CutMix [19]. Mixup [20] linearly interpolates two images and their corresponding labels. Cutout [4] randomly masks out rectangular regions of an image. CutMix [19] cuts a rectangular region from one image and pastes it onto another, while proportionally mixing the labels of both images based on the area of the cut region. CutMix not only preserves the spatial structure in the image but also introduces new contextual information by merging parts of two different images. By maintaining sharper object boundaries and increasing

data diversity, CutMix can improve both accuracy and robustness, especially in scenarios with limited labeled data. We adopt a CutMix-style approach for our augmentation scheme.

Recently, ITMix [7] mixes an image with another image and a text with another text, respectively. In contrast, our method mixes text images with template images. Also, ITMix only focuses on transferring CLIP [14] to image classification while our method enhances a large vision-language model to alleviate the illiteracy problem.

In this paper, we applied image-text mixing to augment synthetic images containing texts, particularly for underrepresented languages. Then, we leverage these images to improve the performance of a Vision Language Model (VLM), MiniCPM-V [17], in recognizing Korean texts within images. By learning diverse augmented training data that merges various text styles and layouts, the VLM can recognize Korean texts in complex and various contexts. Our findings highlight the novel application of the mix augmentation in addressing the issue of data shortage for underrepresented languages and enhancing the multimodal understanding of underrepresented languages in VLMs.

2.2. Vision Language Model

Vision Language Models (VLMs) are designed to integrate different modalities, allowing them to process both images and natural languages. Recent open-source VLMs include MiniCPM-V [17], InternVL [3], and LLaVA [12]. LLaVA [12] is an end-to-end trained large VLM that connects a vision encoder and a large language model for general-purpose visual and language understanding. MiniCPM-V [17] is an efficient VLM that is deployable on end-side devices to achieve a good balance between performance and efficiency. Using techniques such as adaptive visual encoding, multilingual generalization, and the RLAIIF-V method, MiniCPM-V shows comparable performance with GPT-4V and other recent VLMs despite its compact size of 8 billion parameters. For real-world applications, we fine-tune MiniCPM to verify the effectiveness of the proposed method.

3. Proposed Method

The overall pipeline of our image-text mix augmentation scheme is depicted in Fig. 1. Our scheme generates synthetic images, where each synthetic image is obtained by integrating a text image into a patch of a template image representing common textual contexts. Focusing on Korean, a particularly challenging underrepresented language, we introduce a new dataset, the Korean Image-Text Mix dataset. This dataset is composed of synthetic images categorized into four types reflecting real-world scenarios: 1) single-word frontal, 2) multi-word frontal, 3) single-word skewed,

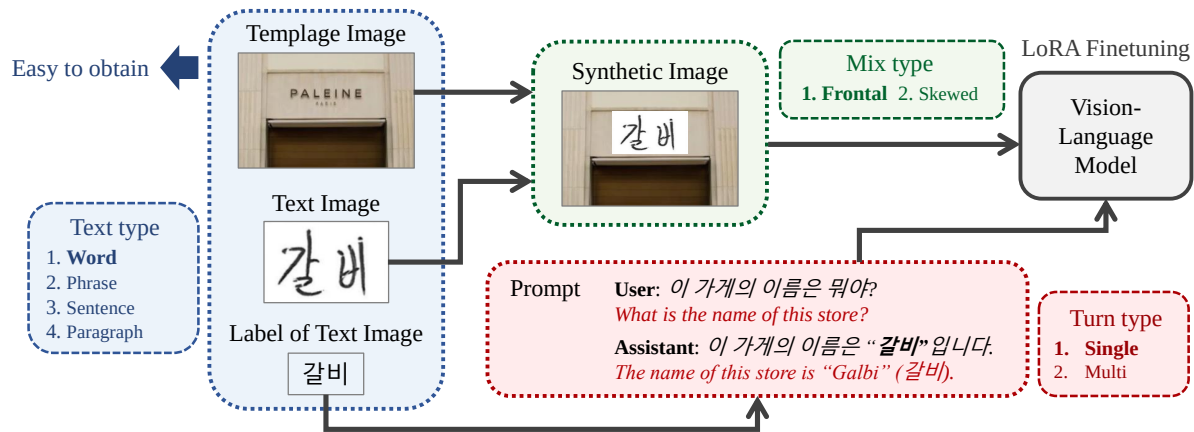


Figure 1: Overall pipeline. We generate a large-scale synthetic dataset by combining text images (words or sentences) into patches of template images representing common textual contexts (receipts, signs, etc.) through our image-text mix augmentation scheme. Labels of the synthetic data are the corresponding texts in the text images. Then, we fine-tune a vision language model using the synthetic data.

and 4) multi-sentence frontal. This dataset is then used to fine-tune a vision language model, MiniCPM-V [17], using low-rank adaptation [8].

3.1. Image-Text Mix Augmentation Scheme

To address the illiteracy problem of VLMs, which means poor performance in recognizing underrepresented language, we propose a novel augmentation scheme that constructs a large-scale realistic dataset containing underrepresented language. As illustrated in Fig. 2a, the proposed scheme generates a synthetic image by combining two images: 1) a text image containing words or sentences that are written in an underrepresented language and 2) a template image representing common textual contexts (e.g. receipts, signs, books, and product labels). For the synthetic image, a patch corresponding to the textual contexts in the template images is selectively removed and replaced with the text image. This process involves identifying the coordinates of the patch of the template image, replacing them with a white background, and subsequently inserting the resized text image to match the designated patch. Then, the label of the synthetic image is set to a natural language sentence that describes texts in the text image. The proposed scheme is categorized into four distinct variations, each applied to different template images to introduce diverse augmentation scenarios. The details and technical configurations for each variation are provided in Sec. 3.2.

3.2. Korean Image-Text Mix Dataset

Among the underrepresented languages, the Korean language presents a unique challenge. Recent open-source VLMs exhibit strong performance in English and Chinese

where resources are abundant, but struggle significantly in Korean. This performance gap highlights the necessity of a large-scale Korean text dataset. Therefore, we propose a new dataset called a Korean Image-Text Mix dataset using the proposed scheme.

Text image. The Korean text images are sourced from the Korean text image dataset¹, which include the 6,000 most frequently used Korean words (based on the National Institute of the Korean Language) and sentences. Text images feature characters written in 50 commonly used fonts, along with handwritten samples from individuals of different genders and age groups. To ensure clarity, all texts are presented on plain white backgrounds, allowing the model to focus on recognizing letter shapes and word patterns.

Template image. The template images contain common textual contexts. These include items such as receipts, signs, books, product labels, and other materials where text is typically present. By utilizing the template images, the model is trained on a diverse range of text appearances and contexts, enhancing its ability to generalize effectively in real-world environments.

Image-text mix augmentation. The synthetic images are augmented by applying our image-text mix augmentation to the Korean text image dataset and a variety of template images. During the augmentation process, we ensure that the text remains clear and readable. To avoid distortion, the aspect ratio of the letters is preserved, allowing the model to learn from natural and undistorted text images. Considering various text types and mix types, the synthetic images are divided into four distinct types (i.e., single-word frontal, multi-word frontal, single-word skewed, and multi-sentence

¹<https://www.aihub.or.kr/aihubdata/data/view.do?currMenu=115&topMenu=100&aihubDataSe=data&dataSetSn=81>



Figure 2: Types of synthetic images.

frontal) to reflect various text recognition challenges.

The **Single-Word Frontal Type** is a basic form of data augmentation where the image is taken with the text facing directly towards the camera and contains a single word, as shown in Fig. 2a. This augmentation focuses on enabling the model to recognize individual words in a straightforward manner. The **Multi-Word Frontal Type** consists of images where multiple words are captured from a frontal perspective, as shown in Fig. 2b. This type of augmentation is designed to help the model handle more complex text compositions compared to the single-word scenario. The **Single-Word Skewed Type** is introduced to enhance the generalization ability of the model by imitating real-world conditions where the camera viewpoint is not pointing directly at the text plane, as shown in Fig. 2c. We apply controlled geometric transformations to generate Skewed text images to simulate these conditions during training. The augmentation process includes the following steps: 1) edge detection: we use the Canny edge detection algorithm [1] to isolate text boundaries from the template image for precise transformations. 2) skew angle detection: through the Hough transform [5], we detect straight lines and calculate the skew angle. Using the calculated skew angle, we apply geometric transformations to create the Skewed image, which is then combined with the template image. The **Multi-Sentence Frontal Type** allows the model to handle

longer sentences and those that include line breaks, ensuring it can accurately comprehend the flow of text, as shown in Fig. 2d.

We pay special attention to sentence boundaries during the image-text mix augmentation. When a sentence exceeds a predefined area, it is split into the next line, maintaining the sentence’s continuity and structure. This technique ensures that line breaks and sentence lengths are preserved in a natural way. For the training dataset, we use 50,000, 10,000, 10,000, and 10,000 samples for Single-Word Frontal Type, Multi-Word Frontal Type, Single-Word Skewed Type, and Multi-Sentence Frontal Type, respectively, resulting in a total of 80,000 training samples. For the testing dataset, we use 20,000 samples of various types.

3.3. Fine-Tuning Vision-Language Model

We fine-tune the MiniCPM-V model [17] to enhance its ability to recognize Korean text from augmented images. For parameter-efficient fine-tuning, we leverage LoRA fine-tuning [8]. Detailed fine-tuning settings are described in Sec. 4.2. In fine-tuning, prompts are specifically tailored to target text extraction from these augmented images. We employ a single-turn prompting approach for Single-Word Frontal Type, Single-Word Skewed Type, and Multi-Sentence Frontal Type, where the text structure is straightforward and can be interpreted in a single inference. For the Multi-Word Frontal type, the single-turn approach is crucial for reading structured sentences or paragraphs. This method ensures the model preserves semantic continuity across line breaks, capturing the full context without disruption. We employ a multi-turn prompting approach for Multi-Word Frontal Type to handle the increased complexity. At each step, the model is guided to focus on specific segments of the image, progressively building a comprehensive understanding of the full context.

Single-turn prompt. Single turn prompts represent straightforward questions that the model might encounter in practical applications, such as identifying text on signs, and labels. For instance, in the case of a single-word image, the model could be prompted with a query like, “What does the sign say?” The model then generates an appropriate response by interpreting the content of the image.

Multi-turn prompt. The implementation of multi-turn prompts with multiple texts allows the model to handle complex interactions. It also ensures that the system can maintain context throughout the dialogue, which is critical for accurately interpreting and responding to queries based on structured documents or lengthy textual information.

4. Experiments

4.1. Datasets

Korean Image-Text Mix dataset. We introduce a new dataset called the Korean Image-Text Mix dataset. More details are described in Sec. 3.2.

Text-In-The-Wild dataset. To evaluate our model, we use the Text-In-The-Wild dataset², focusing on images containing Korean text in various real-world scenarios. For zero-shot evaluation, this dataset is not included in the training to test our model’s generalization capabilities on text recognition tasks. Prompts are created to encourage concise and single-word responses, keeping the evaluation task simple and clear. Additionally, we employ ChatGPT [13] to generate questions, following [10]. This approach leverages large language models, such as ChatGPT, to address issues of dataset imbalance and enhance the diversity of the training data. After generating the questions, a validation process is conducted using ChatGPT to ensure that the questions are appropriately aligned with the context of the dataset. From the questions, our fine-tuned model generates answers, which is free-form based on the findings in [15], where it observes that restricting output formats can degrade performance. Then, we employ ChatGPT to refine the model’s free-form answers into structured outputs.

4.2. Experimental Setting

Evaluation metrics. For calculating error rates, we use Character Error Rate (CER) because CER is more suitable for languages like Korean, where each character conveys substantial semantic meaning. CER is computed as follows:

$$CER = \frac{N_{sub} + N_{del} + N_{ins}}{N}, \quad (1)$$

where N_{sub} is the number of substitutions, N_{del} is the number of deletions, N_{ins} is the number of insertions, and N is the number of characters. Since frequent insertion errors can result in a CER exceeding 100%, which makes the results harder to interpret, we employ the normalization method proposed in [9], which adjusts excessive insertions and ensures that error rates remain within a reasonable range. To focus on the overall recognition accuracy, we use the Character Recognition Rate (CRR), which is calculated by

$$CRR = 1 - CER. \quad (2)$$

Implementation details. We fine-tune the pre-trained MiniCPM-V [17] using LoRA [8] on a single NVIDIA A100 80GB GPU. The learning rate is increased from the default 1e-6 to 1e-5, and the batch size is 2. To improve training stability, bfloat16 is used. The LoRA rank is set to

²<https://www.aihub.or.kr/aihubdata/data/view.do?currMenu=115&topMenu=100&aihubDataSe=data&dataSetSn=81>

Table 1: Zero-shot evaluation on Text-In-The-Wild dataset.

Method	Book	Sentence	Product	Signboard	Trafficsign	Total
Baseline	0.0159	0.0757	0.0204	0.0450	0.2622	0.0830
Ours	0.3012	0.3135	0.3825	0.5201	0.5293	0.4102

Table 2: Performance comparisons on Korean Image-Text Mix test dataset. The best results are represented as bold numbers.

Method	Single	Multi	Sentence	Skewed	Total
Baseline	0.0096	0.0047	0.086	0.0142	0.0164
Ours w/o Single	0.7704	0.4926	0.7631	0.8412	0.6964
Ours w/o Multi	0.9321	0.1891	0.7923	0.9271	0.7021
Ours w/o Sentence	0.7155	0.3237	0.1146	0.7357	0.5430
Ours w/o Skewed	0.9428	0.5022	0.7973	0.9322	0.7993
Ours	0.9745	0.4964	0.8330	0.9608	0.8202

64, and LoRA is applied to the query, key, and value projection modules.

4.3. Comparison with State-of-the-Art Methods

For evaluation, we compare the original MiniCPM-V model (baseline) with the model fine-tuned with our method. As shown in Tabs. 1 and 2, the results show that our method outperforms the baseline. In Tab. 1 for a zero-shot evaluation on the Text-In-The-Wild dataset, our method achieves a high performance gain (**32.72%**) at Total over the baseline, which shows superior generalization capabilities of our method for text recognition tasks. In Tab. 2 for the proposed Korean Image-Text Mix test dataset, our method achieves a high performance gain (**80.38%**) at Total over the baseline. Notable improvements are seen across all evaluation categories, demonstrating the effectiveness of our proposed method in diverse scenarios.

4.4. Ablation Study

We conduct an ablation study to examine the impact of different image-text mix data types. Tab. 2 shows that the most effective approach is to utilize all data types including ‘Single’ (single-word frontal), ‘Multi’ (multi-word frontal), ‘Sentence’ (multi-sentence frontal), and ‘Skewed’ (single-word skewed). Removing the ‘Sentence’ type results in a noticeable performance drop, especially compared to the model trained on the full dataset. This is because images often contain multiple text instances or full sentences, making it challenging for the model to accurately analyze the specific text of interest.

4.5. Qualitative Results

To further evaluate the effectiveness of our data augmentation technique, we conducted a qualitative analysis

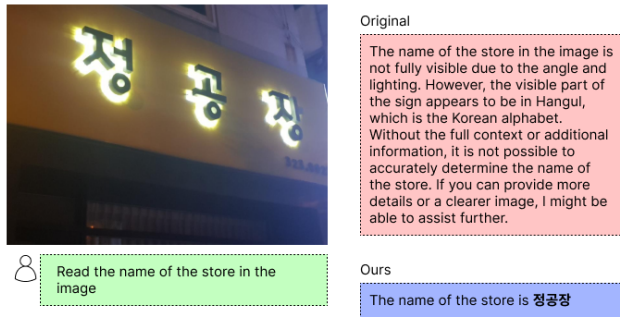


Figure 3: Qualitative result of the original MiniCPM-V and our fine-tuned model.

by applying our model to real-world images containing Korean text. In Fig. 3, our fine-tuned model shows improved recognition of Korean text compared to the original model. The baseline model fails to provide a correct answer, likely due to difficulties with lighting and skewed texts. In contrast, our model correctly identifies the store name. This demonstrates our model’s text recognition capability in real-world conditions, such as challenging lighting conditions and skewed texts.

5. Conclusion

This paper addresses the illiteracy problem of large Vision-Language Models (VLMs) in recognizing underrepresented languages, particularly Korean. To mitigate the lack of training data, we propose a novel image-text mix augmentation scheme and generate a large-scale synthetic Korean Image-Text Mix dataset. The VLM fine-tuned on this dataset significantly improves its Korean text recognition performance, both on our test set and a public real-world dataset. Our results demonstrate the effectiveness of our scheme in creating realistic synthetic data to enhance VLM capabilities in underrepresented languages. Also, the effectiveness of our scheme suggests its potential applicability to other underrepresented languages, such as Tagalog, Arabic, and Turkish. Future work will extend our scheme to additional underrepresented languages.

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